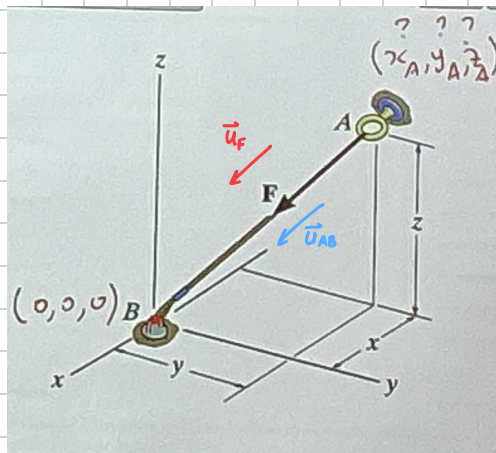


Week 5



AB is 9m long

$$F = \{ 350i - 250j - 450k \} N$$

find (x_A, y_A, z_A)

$$|\vec{r}_{AB}| = 9.0m$$

$$① \vec{u}_F = \frac{\vec{F}}{|\vec{F}|}$$

↓
direction of force F

$$② \vec{u}_{AB} = \frac{\vec{r}_{AB}}{|\vec{r}_{AB}|} ?$$

$$③ \vec{u}_F = \vec{u}_{AB} \rightarrow \text{Solve for } x_A, y_A, z_A$$

$$① \quad \vec{F} = \{350\mathbf{i} - 250\mathbf{j} - 450\mathbf{k}\} \text{ N}$$

$$|\vec{F}| = \sqrt{350^2 + 250^2 + 450^2} = 622.5 \text{ N}$$

$$\vec{U}_F = \frac{\vec{F}}{|\vec{F}|} = \{0.5623\mathbf{i} - 0.4016\mathbf{j} - 0.7229\mathbf{k}\}$$

$$|\vec{r}_{AB}| = 9.0 \text{ m}$$

$$\vec{r}_{AB} = (x_B - x_A)\mathbf{i} + (y_B - y_A)\mathbf{j} + (z_B - z_A)\mathbf{k}$$

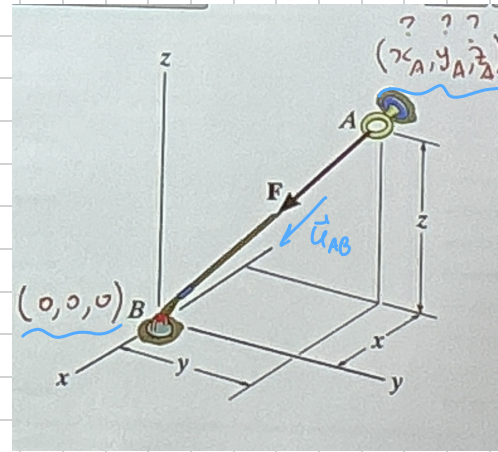
$$= \{-x_A\mathbf{i} - y_A\mathbf{j} - z_A\mathbf{k}\} \text{ m}$$

$$\vec{U}_{AB} = \frac{\vec{r}_{AB}}{|\vec{r}_{AB}|}$$

↳ direction of force

$$\vec{U}_{AB} = \left\{ \frac{-x_A}{9.0}\mathbf{i} - \frac{y_A}{9.0}\mathbf{j} - \frac{z_A}{9.0}\mathbf{k} \right\}$$

$$\vec{U}_F = \vec{U}_{AB}$$



$$\vec{U}_F = 0.5623\mathbf{i} - 0.4016\mathbf{j} - 0.7229\mathbf{k} \rightarrow ①$$

$$\vec{U}_{AB} = \frac{-x_A}{9}\mathbf{i} - \frac{y_A}{9}\mathbf{j} - \frac{z_A}{9}\mathbf{k} \rightarrow ②$$

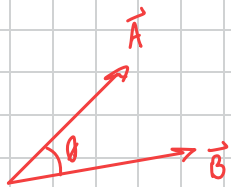
$$-\frac{x_A}{9} = 0.5623 \rightarrow x_A = -5.06 \text{ m} \rightarrow ①$$

$$-\frac{y_A}{9} = -0.4016 \rightarrow y_A = 3.61 \text{ m} \rightarrow ②$$

$$-\frac{z_A}{9} = -0.7229 \rightarrow z_A = 6.51 \text{ m} \rightarrow ③$$

Quiz One! vectors part 4

Dot product



finding Angle from two vectors

Written as $\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta$

↳ scalar quantity output

Laws:

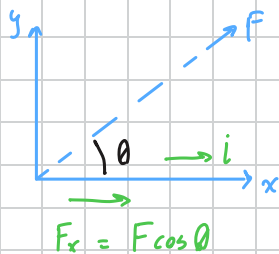
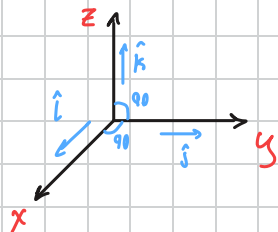
① $A \cdot B = B \cdot A$

② $a(A \cdot B) = (aA) \cdot B = A \cdot aB$

③ $A \cdot (B + D) = A \cdot B + A \cdot D$

$$\hat{i} \cdot \hat{i} = (1)(1) \cos 0^\circ = \underline{\underline{1}} \quad \hat{i} \cdot \hat{j} = (1)(1) \cos 90^\circ = \underline{\underline{0}}$$

$$\hat{i} \cdot \hat{k} = (1)(1) \cos 90^\circ = \underline{\underline{0}}$$



$$\vec{F} \cdot \hat{i} = |\vec{F}| |\hat{i}| \cos \theta$$

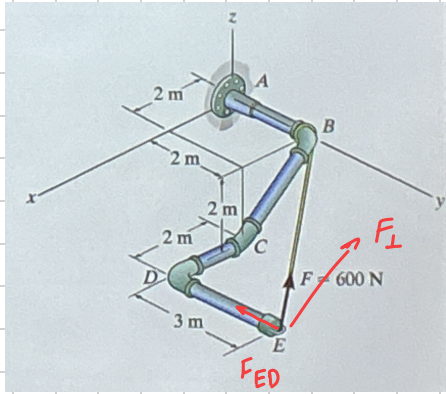
$$F_x = F(1.0) \cos \theta$$

dot product to itself i.e. $\hat{k} \cdot \hat{k} = 1$
to others = 0

$$\begin{aligned} \vec{A} \cdot \vec{B} &= (A_x \hat{i} + A_y \hat{j} + A_z \hat{k}) \cdot (B_x \hat{i} + B_y \hat{j} + B_z \hat{k}) \\ &= (A_x B_x + A_y B_y + A_z B_z) \rightarrow \text{Scalar} \end{aligned}$$

$$\begin{aligned} &A_x B_x \hat{i} \cdot \hat{i} + A_x B_y \hat{i} \cdot \hat{j} + A_x B_z \hat{i} \cdot \hat{k} \\ &+ A_y B_x \hat{j} \cdot \hat{i} + A_y B_y \hat{j} \cdot \hat{j} + A_y B_z \hat{j} \cdot \hat{k} \\ &+ A_z B_x \hat{k} \cdot \hat{i} + A_z B_y \hat{k} \cdot \hat{j} + A_z B_z \hat{k} \cdot \hat{k} \end{aligned}$$

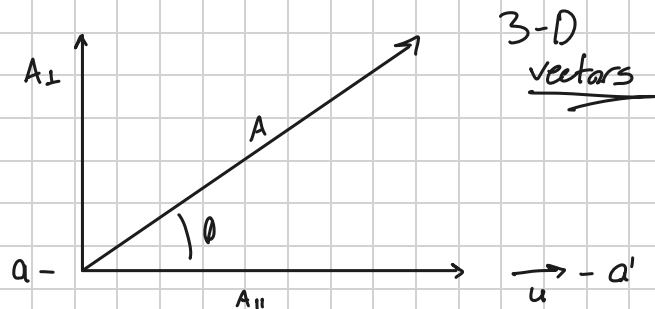
$$\theta = \cos^{-1} \left(\frac{A_x B_x + A_y B_y + A_z B_z}{AB} \right)$$



$$A_{\parallel} = A \cdot u = A \cos \theta$$

$$A_{\parallel} = A \cdot u = Au \cos \theta = (A_x u_x + A_y u_y + A_z u_z)$$

Dot product



$$A_{\parallel} = (A \cdot u) u$$

$\uparrow$ magnitude $\nwarrow$ direction

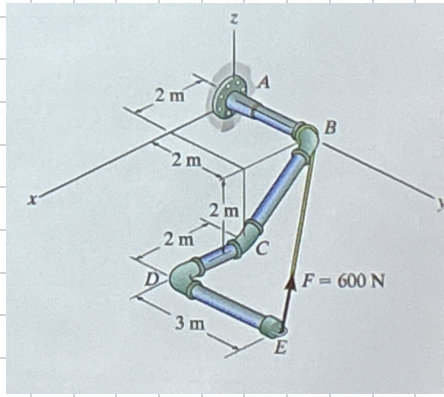
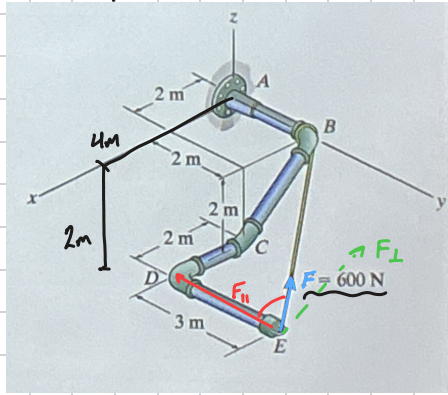
$$A = A_{\parallel} + A_{\perp}$$

$\nearrow$ vector form

A is the resultant
of the two components
all in cartesian form

$$A_{\perp} = \sqrt{A^2 - A_{\parallel}^2}$$

Important question



$$F_{\parallel ED} = \vec{F} \cdot \vec{u}_{ED}$$

$$\textcircled{1} \vec{F} = 600\text{N} \cdot \vec{u}_F$$

$$\vec{u}_F = \vec{u}_{EB}$$

$$\vec{u}_{EB} = \frac{\vec{r}_{EB}}{|\vec{r}_{EB}|} = \checkmark$$

$$\vec{u}_{ED} = \frac{\vec{r}_{ED}}{|\vec{r}_{ED}|}$$

$$|\vec{F}| = 600\text{N}$$

$$\textcircled{1} \vec{F} = 600\text{N} \cdot \vec{u}_{EB}$$

$$\vec{u}_{EB} = \frac{\vec{r}}{|\vec{r}_{EB}|}$$

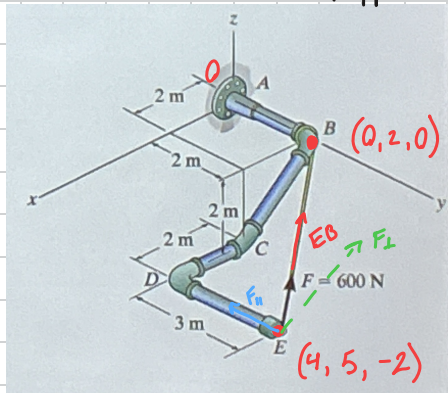
$$\textcircled{2} \vec{u}_{ED} = \frac{\vec{r}}{|\vec{r}_{ED}|} = \{0\hat{i} - \hat{j} + 0\hat{k}\}$$

p.p.c

$$\textcircled{3} F_{\parallel ED} = \vec{F} \cdot \vec{u}_{ED} = \text{magnitude}$$

$$\textcircled{4} F_{\perp ED} = \sqrt{|\vec{F}|^2 - F_{\parallel}^2}$$

Conferim proper cords



②

$$\vec{u}_{ED} = \frac{\vec{r}_{ED}}{|\vec{r}_{ED}|}$$

$$= \frac{\{0\hat{i} - 3\hat{j} + 0\hat{k}\}}{\sqrt{0^2 + 3^2 + 0^2}}$$

$$= \{0\hat{i} - \hat{j} + 0\hat{k}\}$$

③

$$F_{||} = \vec{F} \cdot \vec{u}_{ED} = 334.2 \text{ N}$$

④

$$F_{\perp} = \sqrt{F^2 - F_{||}^2} = \sqrt{600^2 - 334.2^2}$$

$$= 498 \text{ N}$$

①

$$\vec{u}_{EB} = \frac{\vec{r}_{EB}}{|\vec{r}_{EB}|}$$

$$\vec{r}_{EB} = (x_B - x_E)\hat{i} + (y_B - y_E)\hat{j} + (z_B - z_E)\hat{k}$$

$$= (0 - 4)\hat{i} + (2 - 5)\hat{j} + (0 - (-2))\hat{k}$$

$$= \{-4\hat{i} - 3\hat{j} + 2\hat{k}\} \text{ m}$$

$$\vec{u}_{EB} = \frac{\vec{r}_{EB}}{|\vec{r}_{EB}|} = \frac{\{-4\hat{i} - 3\hat{j} + 2\hat{k}\}}{\sqrt{(-4)^2 + (-3)^2 + 2^2}}$$

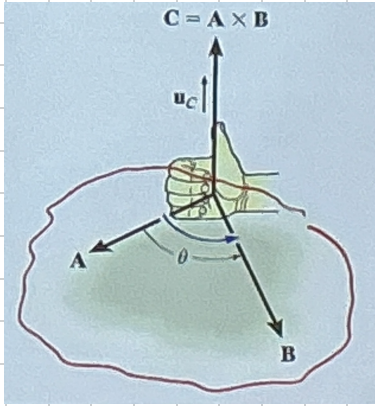
$$= \{-0.7428\hat{i} - 0.5571\hat{j} + 0.3714\hat{k}\}$$

$$\vec{F} = 600 \text{ N} \cdot \vec{u}_{EB}$$

$$= \{-445.6\hat{i} - 334.25\hat{j} + 222.8\hat{k}\} \text{ N}$$

Cross product

$$C = A \times B$$



$$\begin{array}{c} \cancel{x_1} \left| \begin{array}{cc} y_1 & z_1 \\ y_2 & z_2 \end{array} \right| \cancel{z_1} \\ \cancel{z_2} \left| \begin{array}{cc} x_1 & y_1 \\ x_2 & y_2 \end{array} \right| \cancel{x_2} \end{array}$$

$y_1 z_2 + z_1 y_2$
↓
 $(y_1 z_2 + z_1 y_2, y_3 z_3)$

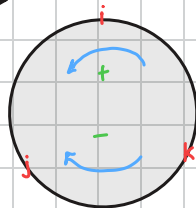
$$A = (x_1, y_1, z_1)$$

$$B = (x_2, y_2, z_2)$$

$$i \times j = (1)(1) \sin 90^\circ = 1$$

$$i \times j = k$$

Trick



$$i \times j = k$$

$$k \times i = -i$$

$$\underline{i \times i = 0}$$

$$i \times k = -j$$

$$j \times i = -k$$

$$j \times j = 0$$

$$j \times k = i$$

$$k \times i = j$$

$$k \times j = -i$$

$$k \times k = 0$$

$$\begin{array}{c} 4 \left| \begin{array}{cc} 1 & 5 \\ 4 & 6 \end{array} \right| 3 \times 8 \\ 3 \left| \begin{array}{cc} 4 & 6 \\ 2 & 9 \end{array} \right| 7 \end{array}$$

$7 + 20$
↓
 $(27, 6, 1)$